Holmenkollen Ski Festival 2026

Device Data Analysis Report

Based on telco device data collected 9–23 March 2026

Executive Summary

This report analyses mobile device data collected across five monitoring areas in Oslo during the 2026 Holmenkollen Ski Festival. The data covers the period from 9 March (pre-event baseline) through 23 March (post-event), capturing two distinct event phases: the FIS World Cup Nordic weekend (14–15 March) and the BMW IBU World Cup Biathlon (19–22 March).

Key finding: The Biathlon weekend generated significantly higher crowd density than the Nordic weekend, with the Holmenkollen Ski Arena peaking at 5,657 devices on Saturday 21 March (Pursuit day) compared to a peak of 3,630 during the Nordic 50km on 14 March. The Nordic weekend was heavily impacted by fog and poor visibility, while the Biathlon weekend enjoyed warm spring weather.

International presence: During the Biathlon weekend, 61.5% of devices at the Holmenkollen Ski Arena were international (non-Norwegian SIMs), led by Germany, France, and Sweden. The Nordic weekend saw 52% international at the arena. Across all five areas, international devices accounted for 33.7% of all readings during Biathlon week.

Monitoring Areas

Device counts were collected at approximately 5-minute intervals from cell tower connections within five defined geographic polygons:

Area	Role
Oslo Central Station	Primary transit hub and baseline indicator for crowd influx into Oslo
Majorstua Metro Station	Key metro interchange and choke point for Holmenkollen Line 1 access
Holmenkollen Ski Arena	Ticketed stadium zone — finish line, grandstands, commercial areas
Holmenkollen Marka Arena	Forest area north/west of stadium — trails, camping, informal spectating
Sognsvann Arena	Northern access point via Metro Line 5 and cross-country ski trail to Holmenkollen

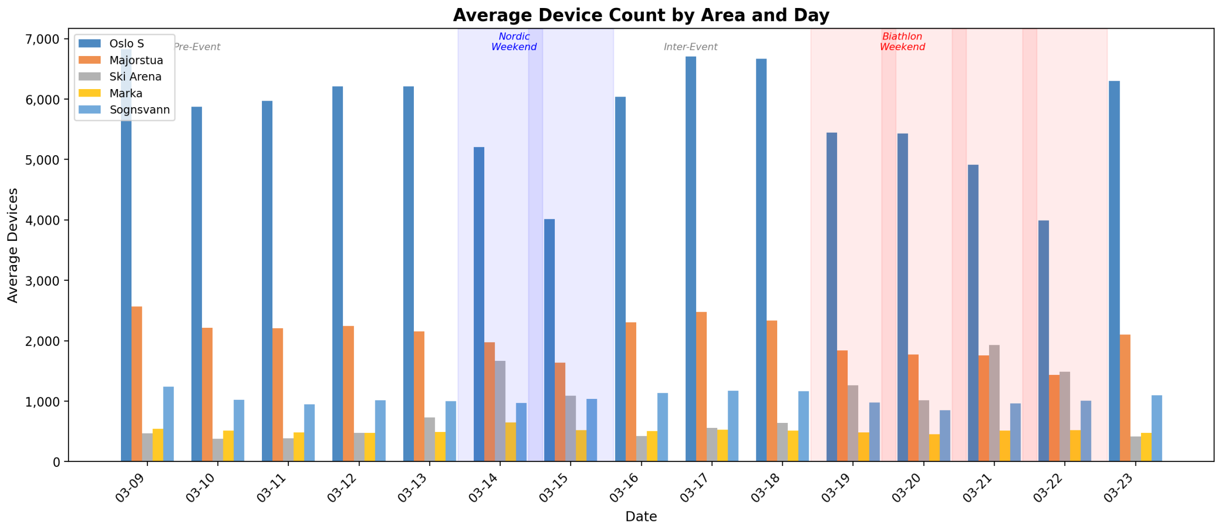
Event Calendar

Phase	Dates	Events / Context
Pre-Event Buffer	9–13 March	Normal weekday baselines; event setup underway at Holmenkollen from 12–13 March
Nordic Weekend	14–15 March	Historic Double 50km (men + women), ski jumping, Nordic Combined PCR. Heavy fog, rain, poor visibility.
Inter-Event Buffer	16–18 March	Normal weekdays; baselines reset. Venue changeover to biathlon configuration.

Biathlon Weekend	19–22 March	Sprint (Thu/Fri), Pursuit (Sat), Mass Start (Sun). Clear skies, warm temperatures up to 14°C.
Post-Event	23 March	Monday return to baseline.

Daily Device Overview

The chart below shows the average device count per area for each day of the monitoring period. The shaded blue region marks the Nordic weekend, and the shaded red region marks the Biathlon weekend.



Oslo Central Station and Majorstua both show reduced device counts on event days (particularly weekends), reflecting their role as transit nodes rather than destinations — visitors pass through rather than linger. Conversely, the Holmenkollen Ski Arena shows sharp spikes on event days, with the Biathlon Saturday (21 March) producing the highest peak of any day across the entire period.

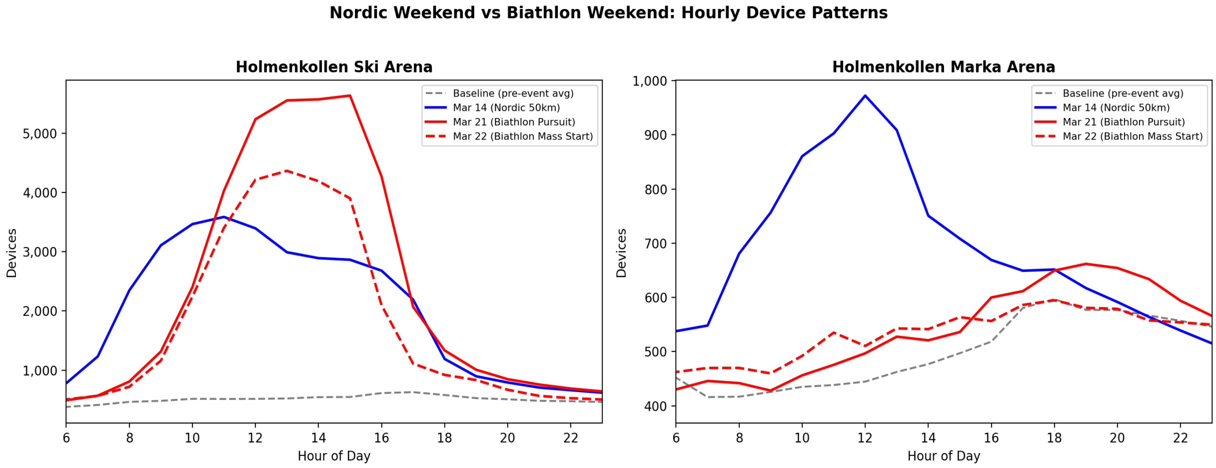
Key Metrics by Area

Area	Baseline Avg	Nordic Avg	Nordic Peak	Biathlon Avg	Biathlon Peak
Oslo S	8,289	4,609	8,278	4,945	9,086
Majorstua	2,831	1,807	2,973	1,700	2,555
Ski Arena	543	1,377	3,630	1,422	5,657
Marka	483	581	1,023	491	694
Sognsvann	1,024	1,002	1,235	949	1,213

Baseline averages are calculated from weekday daytime hours (08:00–18:00) during the pre-event period (9–13 March).

Nordic vs Biathlon: Venue Patterns

The two event weekends produced strikingly different crowd patterns at the Holmenkollen venues. The charts below compare hourly device counts at the Ski Arena and Marka Arena on key event days against the pre-event baseline.

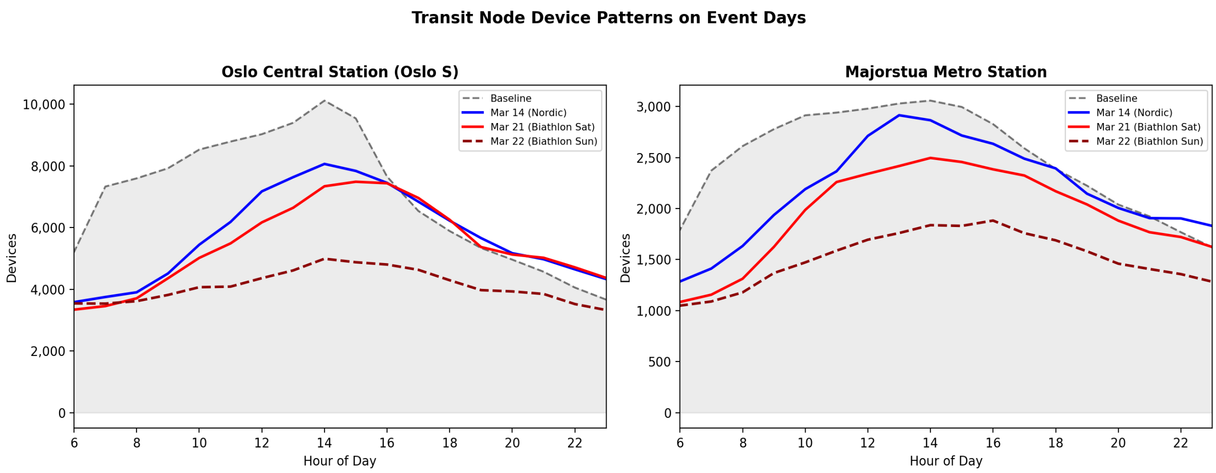


Holmenkollen Ski Arena: The Biathlon Pursuit day (21 March) peaked at 5,657 devices around 15:30, a 56% increase over the Nordic 50km peak of 3,630 on 14 March. The Biathlon Mass Start (22 March) also surpassed the Nordic day with a peak of 4,409. This contrast reflects both the weather impact (fog suppressed Nordic attendance) and the nature of biathlon as a tightly-packed stadium sport versus the dispersed 50km trail format.

Marka Arena: The forest area showed the opposite pattern. The Nordic weekend (14 March) generated the Marka's highest readings of the entire period, peaking at 1,023 devices as fans spread along the 50km trail. During the Biathlon weekend, Marka device counts remained near baseline levels, confirming that biathlon crowds concentrated inside the stadium perimeter.

Transit Node Patterns

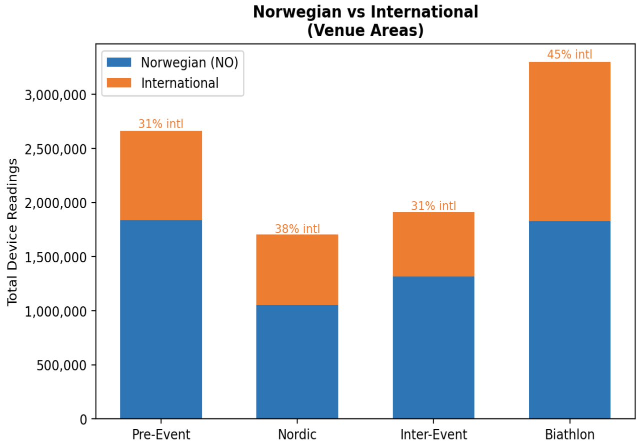
Oslo Central Station and Majorstua Metro Station serve as upstream indicators of crowd flow toward Holmenkollen. Their device patterns on event days reveal how spectators moved through the city.

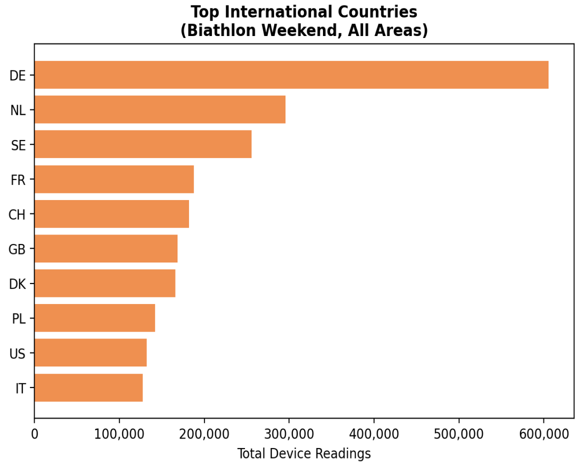


Both transit nodes show lower-than-baseline readings on event Saturdays and Sundays (14–15 March and 21–22 March). This is expected: these are weekend days with reduced commuter traffic, and the festival crowds represent a smaller absolute number than the weekday commuter volume. The Biathlon Sunday (22 March) shows the lowest Majorstua readings, consistent with reported metro line closures (Fornebubanen integration works) that forced spectators onto substitute bus services and alternative routes via Sognsvann.

International Visitor Analysis

Using MCC (Mobile Country Code) data from device SIM registrations, we can estimate the share of Norwegian versus international visitors across the monitoring areas.





Venue areas during Biathlon weekend: International devices accounted for 61.5% of all readings at the Holmenkollen Ski Arena during the Biathlon weekend, up from 52.0% during the Nordic weekend. Germany was the dominant international presence with over 323,000 device readings, followed by France (116,000) and Sweden (76,000). This reflects the strong European biathlon fanbase ahead of the 2026 Milano Cortina Olympics.

Overall trend: Across all five areas, the international share rose from 24.5% during the pre-event period to 33.7% during the Biathlon weekend, indicating a significant influx of foreign visitors specifically for the biathlon events.

Top International Countries — Biathlon Weekend

Country	Device Readings	Context
Germany (DE)	323,079	Largest international contingent; strong biathlon tradition
France (FR)	116,222	Significant presence driven by competitive athletes and fans
Sweden (SE)	75,792	Neighbouring country; strong cross-border attendance
Denmark (DK)	58,525	Scandinavian neighbours
Netherlands (NL)	48,989	Notable given distance; reflects winter sport enthusiasm

Key Takeaways

- **Biathlon outperformed Nordic in crowd density:** The Ski Arena peak on Biathlon Saturday (5,657) was 56% higher than the Nordic 50km peak (3,630), driven by better weather and the stadium-centric format.
- **Weather was the decisive factor for Nordic attendance:** Dense fog on 14–15 March degraded the spectator experience and suppressed arena attendance well below what a historic Double 50km would normally attract.
- **Strong international draw for biathlon:** 61.5% of Ski Arena devices during biathlon were international, with Germany, France, and Sweden as the top three countries.
- **Marka usage diverged by event type:** The forest area saw its highest activity during the 50km Nordic event (fans along trails) but returned to baseline during biathlon, which is inherently stadium-focused.
- **Transit infrastructure under pressure:** The metro closures on 21–22 March (Fornebubanen works) coincided with peak biathlon attendance, with Majorstua showing its lowest readings of the period, suggesting crowds were diverted to alternative routes or surface-level transport.

Methodology

Device counts are based on mobile devices connected to cell towers within defined geographic polygons, sampled approximately every 5 minutes. Counts represent concurrent connected devices, not unique visitors — a single person may be counted in multiple consecutive readings. International visitor estimates use the Mobile Country Code (MCC) of the device's SIM registration. The data was collected by the commercial team's realtime monitoring platform.

Report generated 23 March 2026